

Prateek Saxena
Associate Professor of Computer Science
National University of Singapore
prateeks@comp.nus.edu.sg

March 14, 2026

Faculty Affairs
Vector Institute
Toronto, Ontario

Dear Vector Institute Faculty Affairs Committee,

I am writing to apply for membership as a Vector Institute Faculty Member. I am an Associate Professor with tenure in the Department of Computer Science at the National University of Singapore and a nominee for the Canada Impact+ Research Chair at the University of Toronto Mississauga in Mathematical and Computational Sciences. I am enthusiastic about contributing to Vector’s research community, particularly in ML security—a field I have helped shape over the past decade.

My ML security research has been foundational. I was among the first to demonstrate *data poisoning* attacks in neural network training (ACSAC 2016); this work is cited in the OWASP ML Security Top Ten. I served as Area Chair for “Machine Learning Security” at ACM CCS 2020–21, and have developed formal ML security abstractions—geometric characterisations, causal models, and post-quantum formulations for model inversion. Most recently, my group achieved the first computationally tractable defences against data poisoning with provable guarantees. I am extending this toward AI-assisted migration to secure languages (including C-to-Rust for large codebases) and secure hardware abstractions (CAPSTONE).

Beyond ML security, my work has had substantial real-world impact. I designed the security architecture for the Google Chrome extensions platform (deployed 2010, billions of users), language-based defences for cross-site scripting that formed the basis of the Closure compiler and Trusted Types, and web sandboxing integrated into Dropbox. In decentralized systems, I introduced practical blockchain sharding (Elastico, CCS 2016), co-founded Zilliqa (billion-dollar valuation, 2018), and established DeFi smart-contract auditing with Oyente and MAIAN. The “verifier dilemma” (CCS 2015) received the ACM CCS Test of Time Award (2025). I directed NUS’s Crystal Center for blockchain research (2018–2025), and my PhD graduates hold faculty positions at Georgia Tech and ETH Zurich. Recognitions include MIT TR35 “Top 10 Innovators” Asia (2017) and the Google Security & Privacy Researcher Award (2018).

As a Vector Faculty Member, I plan to advance ML security along three lines: training-free defences for adversarial and privacy-leaking outputs; secure reasoning and information flow for AI agents; and verifiable AI-assisted migration to secure languages. I look forward to collaborating with Vector researchers and contributing to Canada’s leadership in trustworthy AI.

Thank you for considering my application. I look forward to the possibility of joining the Vector Institute.

Sincerely,

Prateek Saxena